



Province of the
EASTERN CAPE
EDUCATION



**NATIONAL SENIOR
CERTIFICATE/*NASIONALE SENIOR
SERTIFIKAAT***

GRADE/*GRAAD* 11

NOVEMBER 2022

**TECHNICAL MATHEMATICS P2/*TEGNIесе WISKUNDE V2*
MARKING GUIDELINE/*NASIENRIGLYN***

MARKS/*PUNTE*: 150

This marking guideline consists of 16 pages./
Hierdie nasienriglyn bestaan uit 16 bladsye.

NOTE:

- Continuous accuracy (CA) applies only where indicated in this marking guideline.
- Assuming values/answers in order to solve a problem is unacceptable.

LET WEL:

- *Volgehoue akkuraatheid (CA) is slegs van toepassing soos in hierdie nasienriglyn aangedui.*
- *Aanvaarding van waardes/antwoorde om 'n probleem op te los, is onaanvaarbaar.*

MARKING CODES / NASIENKODES	
M	Method / Metode
A	Accuracy / Akkuraatheid
AO	Answer only / Slegs antwoord
CA	Consistent accuracy / Deurlopende akkuraatheid
F	Formula / Formule
I	Identity / Identiteit
R	Rounding / Afronding
S	Simplification / Vereenvoudiging
ST	Statement / Bewering
RE	Reason / Rede
ST RE	Statement and correct reason / Bewering en korrekte rede
SF	Substituting correctly in correct formula / Korrekte vervanging in die korrekte formule
NPU	No penalty for omitting units / Geen penalisering vir eenhede uitgelaat

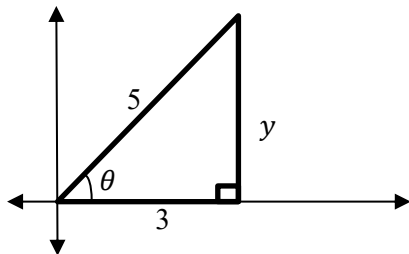
QUESTION/VRAAG 1

Q		M
1.1	$K(4;0)$	✓ 5 ✓ 0 (2)
1.2	$M_{PM} = \left(\frac{x_2 + x_1}{2}; \frac{y_2 + y_1}{2} \right)$ $M_{PM} = \left(\frac{x_M + 4}{2}; \frac{y_M + 3}{2} \right)$ $\therefore 1 = \frac{x_M + 4}{2}$ and/en $\therefore 1 = \frac{y_M + 3}{2}$ $2 = x_M + 4$ and/en $2 = y_M + 3$ $-2 = x_M$ and/en $-1 = y_M$ $\therefore M(-2;-1)$	✓F $\checkmark 1 = \frac{x_M + 4}{2}$ $\checkmark 1 = \frac{y_M + 3}{2}$ ✓Simplification/Ver-eenvoudiging $\checkmark -2 = x_M$ $\checkmark -1 = y_M$ (6)
1.3	$m_{PM} = \frac{y_2 - y_1}{x_2 - x_1}$ $m_{PM} = \frac{3+1}{4+2}$ OR / OF $m_{PM} = m_{QM} = \frac{3-1}{4-2}$ $m_{PM} = \frac{4}{6}$ $m_{PM} = \frac{2}{3}$ OR /OF $m_{PM} = m_{QM} = \frac{2}{3}$	✓F ✓SF ✓CA (3)
1.4	$\tan \hat{PSK} = \frac{2}{3}$ $\hat{PSK} = \tan^{-1} \left(\frac{2}{3} \right)$ $\hat{PSK} = 33,69 \dots^\circ$ $\therefore \hat{SPK} = 180^\circ - 90^\circ - 33,69 \dots^\circ$ $\hat{SPK} = 56,31^\circ$ (Int. $\angle$'s of $\triangle SPK$ /Binne $\angle$ 'e van $\triangle SPK$)	✓M ✓CA ✓M ✓ST CA ✓R (5)

1.5	$\sin \widehat{PSK} = \frac{PK}{PS}$ $\sin 33,69\dots^\circ = \frac{3}{PS}$ $PS \times \sin 33,69\dots^\circ = 3$ $PS = \frac{3}{\sin 33,69\dots^\circ}$ $PS = 5,41 \text{ units/eenhede}$ OR/OF $y = \frac{2}{3}x + c$ $3 = \frac{2}{3}(4) + c \text{ or/of } 1 = \frac{2}{3}(1) + c \text{ or/of } -1 = \frac{2}{3}(-2) + c$ $3 = \frac{8}{3} + c \text{ or/of } 1 = \frac{2}{3} + c \text{ or/of } -1 = \frac{-4}{3} + c$ $\frac{1}{3} = c$ $\therefore y = \frac{2}{3}x + \frac{1}{3}$ $\therefore S(x;0):0 = \frac{2}{3}x + \frac{1}{3}$ $-\frac{1}{3} = \frac{2}{3}x$ $-\frac{1}{2} = x$ $\therefore S\left(-\frac{1}{2}; 0\right)$ $\therefore PS = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $PS = \sqrt{\left(-\frac{1}{2} - 4\right)^2 + (0 - 3)^2}$ $PS = 5,41 \text{ units/eenhede}$	✓M CA from/van 1.4 ✓$\frac{3}{\sin 33,69\dots^\circ}$ ✓CA ✓M ✓CA ✓Answer/antwoord CA (3)
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1.6	$TN: m = \frac{2}{3} (TN \parallel MP)$ $\therefore y = \frac{2}{3}x + c$ $(-1;7): 7 = \frac{2}{3}(-1) + c$ $7 = \frac{-2}{3} + c$ $\frac{23}{3} = c$ $\therefore y = \frac{2}{3}x + \frac{23}{3}$ $\therefore N$ is a point of intersection/is 'n snypunt $\therefore -3x + 15 = \frac{2}{3}x + \frac{23}{3}$ $-3x - \frac{2}{3}x = \frac{23}{3} - 15$ $-\frac{11}{3}x = -\frac{22}{3}$ $x = 2$ $\therefore y = -3(2) + 15$ $y = 9$ $\therefore N(2;9)$	$\checkmark$ ST RE $\checkmark \frac{23}{3} = c$ $\checkmark y = \frac{2}{3}x + \frac{23}{3}$ $\checkmark$ M $\checkmark x = 2$ $\checkmark y = 9$ $\checkmark N(2;9)$ (7)
		[26]

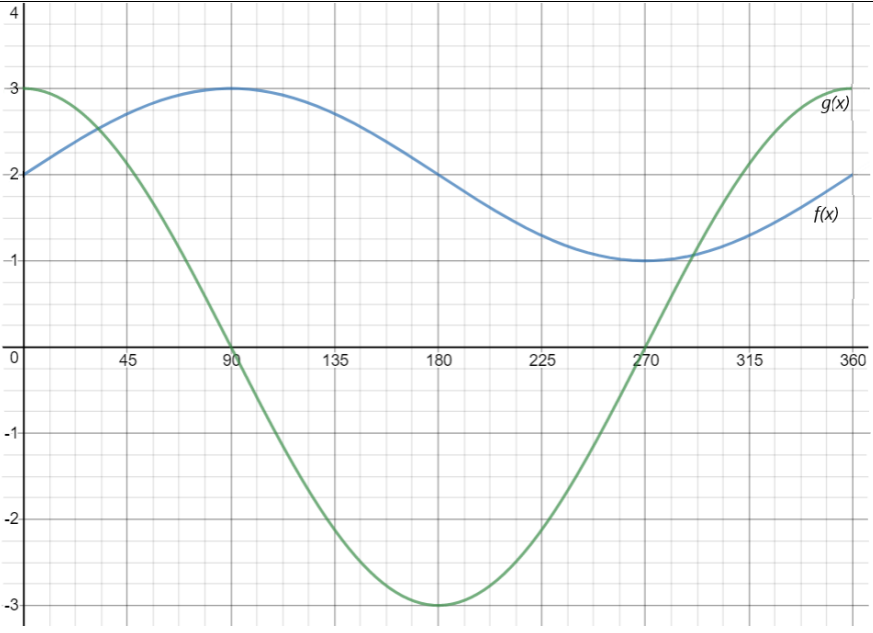
QUESTION/VRAAG 2

Ques		M
2.1.1	41,54	✓A (1)
2.1.2	8,68	✓✓A (2)
2.2.1	$5\cos\theta = 3$ $\cos\theta = \frac{3}{5}$ $\therefore \sin\theta.\sec\theta = \frac{4}{5} \times \frac{5}{3}$ $= \frac{4}{3}$  $(5)^2 = (3)^2 + (y)^2$ $\therefore y = 4$	✓ $\cos\theta = \frac{3}{5}$ ✓ Diagram in correct quadrant/diagram in korrekte kwadrant ✓ $y = 4$ ✓ $\frac{4}{5} \times \frac{5}{3}$ ✓ CA (5)
2.2.2	$\frac{\tan\theta}{\cot\theta} = \frac{\frac{4}{3}}{\frac{3}{4}}$ $= \frac{4}{3} \times \frac{4}{3}$ $= \frac{16}{9}$	✓✓ SF CA ✓ CA (3)
2.2.3	$\cos\theta = \frac{3}{5}$ $\theta = \cos^{-1}\left(\frac{3}{5}\right)$ $\theta = 53,13^\circ$	✓ Using any correct ratio/Maak gebruik van enige korrekte verhouding ✓ M ✓ CA (3)
2.3	$3\sin\theta = -1,026$ $\sin\theta = -0,342$ Ref/Verw $\angle \approx 20^\circ$ $\theta = 180^\circ + 20^\circ$ or/of $360^\circ - 20^\circ$ $\theta = 200^\circ$ or/of 340°	✓ S A ✓ Ref/Verw $\angle$ CA ✓ $180^\circ +$ A ✓ $360^\circ -$ A ✓ S CA (5)
		[19]

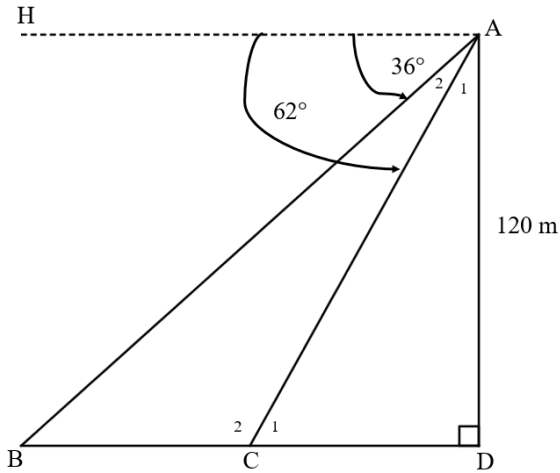
QUESTION/VRAAG 3

Ques		M
3.1.1	$1 - \sin^2 x = \cos^2 x$	$\checkmark \cos^2 x$ (1)
3.1.2	$\sec^2 x - \tan^2 x = 1$	$\checkmark 1$ (1)
3.2	$\frac{\sin(\pi - \theta) \cdot \tan \theta \cdot \sin 270^\circ}{\cos(2\pi - \theta) \cdot \tan(\pi - \theta)} = \frac{(\sin \theta)(\tan \theta)(-1)}{(\cos \theta)(-\tan \theta)}$ $= \tan \theta$	$\checkmark \sin \theta$ A $\checkmark -1$ A $\checkmark \cos \theta$ A $\checkmark -\tan \theta$ A $\checkmark \tan \theta$ CA (5)
3.3	$\text{LHS/LK} = \sin x + \cot x \cdot \cos x$ $= \sin x + \frac{\cos x}{\sin x} \cdot \cos x$ $= \frac{\sin^2 x + \cos^2 x}{\sin x}$ $= \frac{1}{\sin x}$ $= \operatorname{cosec} x$ $= \text{RHS/RK}$	$\checkmark \frac{\cos x}{\sin x}$ I A $\checkmark \frac{\sin^2 x + \cos^2 x}{\sin x}$ $\checkmark 1$ I A $\checkmark \frac{1}{\sin x} = \operatorname{cosec} x$ (4)
		[11]

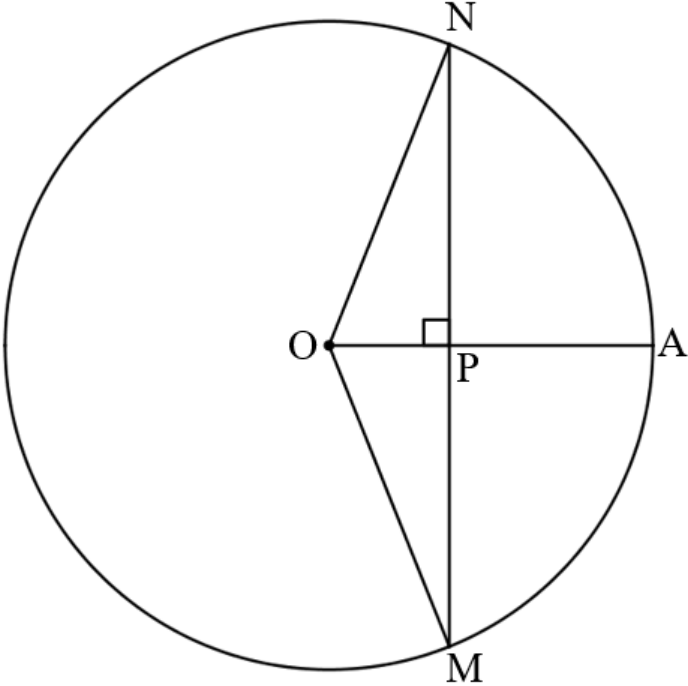
QUESTION/VRAAG 4

Ques		M
4.1	 Learners can use any interval / Leerder kan enige interval gebruik.	<i>g:</i> ✓ cos start and end point/cos begin- en eindpunte A ✓ cos turningpoints/cos draaipunte A ✓ cos x-intercepts/cos x-afsnitte A <i>f:</i> ✓ sin start and end point/sin begin- en eindpunte A ✓ sin turningpoints/sin draaipunte A ✓ sin x-intercepts/sin x-afsnitte A (6)
4.2.1	3	✓ A (1)
4.2.2	$90^\circ \leq x \leq 270^\circ$	✓ $90^\circ \leq$ CA ✓ $\leq 270^\circ$ CA (2)
		[9]

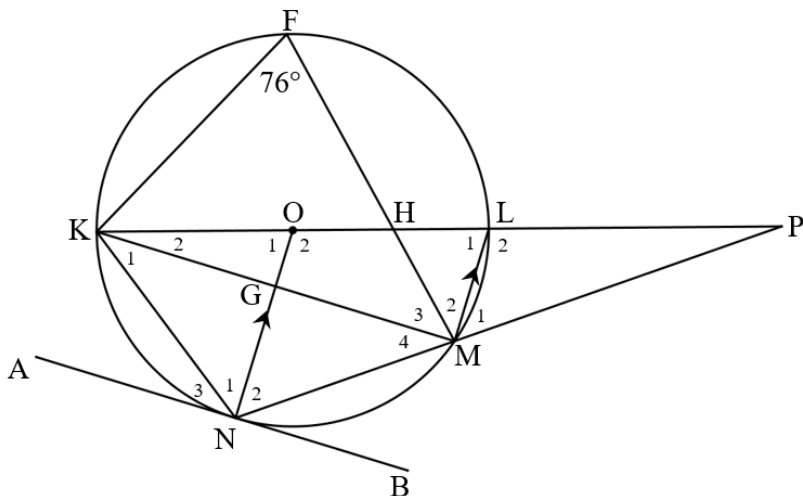
QUESTION/VRAAG 5

Ques		M
		
5.1	$\hat{B} = 36^\circ$ (alt. $\angle$'s = /verw. $\angle$ 'e =; HA $\parallel$ BD)	✓ST ✓RE (2)
5.2	$\hat{C}_1 = 62^\circ$ (alt. $\angle$'s = /verw. $\angle$ 'e =; HA $\parallel$ BD)	✓ST RE (1)
5.3	$\frac{AC}{120} = \operatorname{cosec} 62^\circ$ $AC = \frac{120}{\sin 62^\circ}$ $\approx 135,91 \text{ m}$	✓M ✓S ✓CA (3)
5.4	$\frac{BC}{\sin A_2} = \frac{AC}{\sin B}$ $\frac{BC}{\sin 26^\circ} = \frac{135,91}{\sin 36^\circ}$ $BC = \frac{135,91 \sin 26^\circ}{\sin 36^\circ}$ $\approx 101,36 \text{ m}$	✓F ✓SF CA ✓ $\hat{A}_2 = 26^\circ$ A ✓S ✓CA (5)
5.5	$\text{Area of } \triangle ABC = \frac{1}{2} BC \cdot AC \sin C_2$ $= \frac{1}{2} \times 101,36 \times 135,91 \sin 118^\circ$ $= 6081,67 \text{ m}^2$	✓F ✓SF CA ✓CA (3)
		[14]

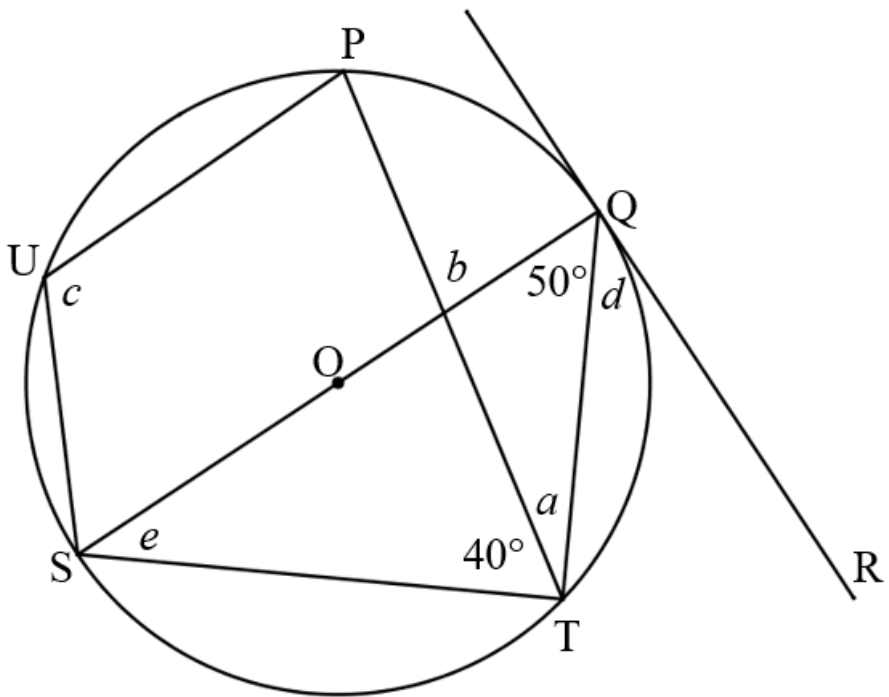
QUESTION/VRAAG 6

Ques		M
6.1	Bisects the chord/ halveer die koord.	✓A (1)
6.2		
	$MP = PN = \frac{48}{2} = 24$ units/eenhede (line from centre $\perp$ to chord / loodlyn uit midpt O na koord) $ON = OM = OA$ (radii) $\therefore ON^2 = OP^2 + PN^2$ (Pyth) $ON^2 = (7)^2 + (24)^2$ $ON^2 = 625$ $ON = \pm\sqrt{625}$ $ON = \pm 25$ $\therefore ON = 25$ units/eenhede $\therefore PA = OA - OP$ $PA = 25 - 7$ $PA = 18$ units/eenhede	✓ST ✓RE ✓ST RE ✓M ✓CA ✓M ✓CA (7)
		[8]

QUESTION/VRAAG 7

Ques		M
7.1	90°	✓A (1)
7.2		
7.2.1	$\hat{L}_1 = 76^\circ$ ($\angle$'s in same segment/ $\angle$ 'e in dieselfde segment)	✓ST ✓RE (2)
7.2.2	$\hat{O}_2 = 76^\circ$ (corresp. $\angle$'s/ooreenkomstige $\angle$ 'e; $ON \parallel LM$)	✓ST ✓RE (2)
7.2.3	$\hat{M}_4 = 38^\circ$ ($\angle$ at centre = 2 x $\angle$ at circumf./midpts $\angle$ = 2 x omtreks $\angle$)	✓ST ✓RE (2)
7.2.4	$\hat{N}_3 = 38^\circ$ (tan chord/raaklyn koord)	✓ST ✓RE (2)
7.2.5	$\hat{N}_1 + \hat{N}_2 = 104^\circ$ (opp $\angle$'s of a cq/teenoorst $\angle$ 'e van 'n kvh)	✓ST ✓RE (2)
7.2.6	$\hat{N}_1 + \hat{N}_3 = 90^\circ$ (rad $\perp$ tan/rad $\perp$ raaklyn)	✓ST ✓RE (2)
7.2.7	$\hat{M}_2 + \hat{M}_3 = 90^\circ$ ($\angle$ in semi-circle/ $\angle$ in semi-sirkel)	✓ST ✓RE (2)
7.2.8	$\hat{M}_1 = 52^\circ$ ($\angle$'s on str. line/ $\angle$ 'e op reguitlyn)	✓ST ✓RE (2)
		[17]

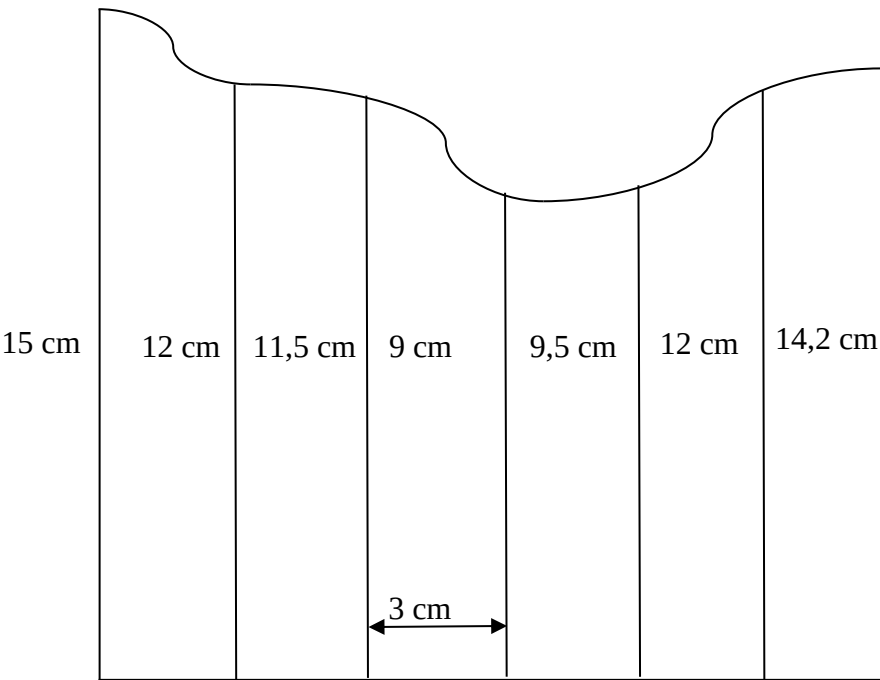
QUESTION/VRAAG 8

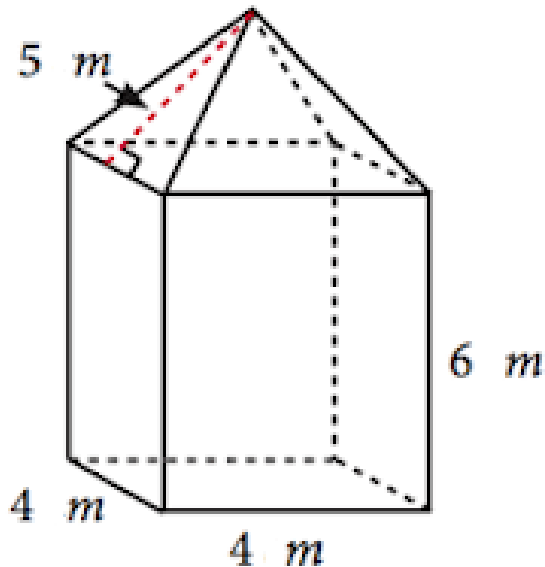
Ques		M
8.1	Equal/ <i>gelyk</i>	✓A (1)
8.2		
8.2.1	$a = 50^\circ$ ($\angle$ in semi-circle/ $\angle$ in semi-sirkel)	✓ST ✓RE (2)
8.2.2	$b = 100^\circ$ (ext $\angle$ of Δ /buite $\angle$ van Δ)	✓ST ✓RE (2)
8.2.3	$c = 140^\circ$ (opp $\angle$'s of a cq/teenoorst $\angle$ 'e van 'n kvh)	✓ST ✓RE (2)
8.2.4	$d = 40^\circ$ (rad $\perp$ tan/rad $\perp$ raaklyn)	✓ST ✓RE (2)
8.2.5	$e = 40^\circ$ (tan chord/raaklyn koord)	✓ST ✓RE (2)
		[11]

QUESTION/VRAAG 9

Ques		M
9.1.1	Area of a sector Opp. van 'n segment = $\frac{r^2\theta}{2}$ $\therefore 8,5 = \frac{(2,1)^2\theta}{2}$ $17 = 4,41\theta$ $3,8548 \dots = \theta$ $\therefore 3,8548 \dots \times \frac{180^\circ}{\pi} = \theta$ $221^\circ = \theta$	✓ F ✓ SF ✓ 3,8545... ✓ 221° (4)
9.1.2	$s = r\theta$ $= (2,1)(3,8548 \dots)$ $= 8,10 \text{ cm}$	✓ F ✓ SF ✓ 8,10 cm (3)
9.2.1	$v = \pi Dn$ $v = \pi(80)(21)$ $v = 1680\pi$ <i>or/of</i> 5278 mm/s	✓ F ✓ SF ✓ 1680π <i>or/of</i> 5278 mm/s (3)
9.2.2	$\omega = 2\pi n$ $\omega = 2\pi(21)$ $\omega = 42\pi$ <i>or/of</i> 132 rad/s OR/OF $v = \omega r$ $\omega = \frac{v}{r}$ $= \frac{1680\pi}{40}$ $= 42\pi \text{ rad/s}$	✓ F ✓ SF ✓ 42π <i>or/of</i> 132 rad/s OR/OF ✓ F ✓ SF ✓ 42π <i>or/of</i> 132 rad/s (3)

QUESTION/VRAAG 10

Ques.		M
		
10.1	$A_T = a(m_1 + m_2 + m_3 + \dots + m_n)$ $A_T = 3(13,5 + 11,75 + 10,25 + 9,25 + 10,75 + 13,1)$ $A_T = 3(68,6)$ $A_T = 205,8 \text{ cm}^2$ OR/OF $A_T = a\left(\frac{o_1 + o_n}{2} + o_2 + o_3 + o_4 + \dots + o_{n-1}\right)$ $A_T = 3\left(\frac{15 + 14,2}{2} + 12 + 11,5 + 9 + 9,5 + 12\right)$ $A_T = 3(68,6)$ $A_T = 205,8 \text{ cm}^2$	✓ F ✓ SF ✓ S ✓ 205,8 cm² ✓ F ✓ SF ✓ S ✓ 205,8 cm² (4)

10.2		
10.2.1	$A_T = (2lh + 2wh) + (2bs + b^2)$ $A_T = (2(4)(6) + 2(4)(6)) + (2(4)(5) + (4)^2)$ $A_T = 96 + 56$ $A_T = 152 \text{ m}^2$	✓ $(2lh + 2wh)$ ✓ $(2bs + b^2)$ ✓ SF ✓ 152 m^2 (4)
10.2.2	$\text{Area of base} = s^2 = (4)^2 = 16 \text{ m}^2$ $(5)^2 = H^2 + \left(\frac{4}{2}\right)^2$ $25 = H^2 + 4$ $21 = H^2$ $\pm\sqrt{21} = H$ $\therefore H = \sqrt{21}$ $\text{Vol} = (lwh) + \left(\frac{1}{3}AH\right)$ $\text{Vol} = (4 \times 4 \times 6) + \left(\frac{1}{3} \times 16 \times \sqrt{21}\right)$ $\text{Vol} = 96 + 24,44 \dots$ $\text{Vol} = 120,44 \text{ m}^3$	✓ 16 m^2 ✓ $(5)^2 = H^2 + \left(\frac{4}{2}\right)^2$ ✓ $\sqrt{21} \text{ CA}$ ✓ F ✓ SF ✓ $120,44 \text{ m}^3 \text{ CA}$ (6)
		[14]
	TOTAL/TOTAAL:	150